ASSESSMENT COVERSHEET

Attach this coversheet as the cover of your submission. All sections must be completed.

Section A: Submission Details

Programme	:	BACHELOR OF COMPUTER ENGINEERING TECHNOLOGY (COMPUTER SYSTEMS) WITH HONOURS
Course Code & Name	:	IPB 49906 FINAL YEAR PROJECT 2
Course Lecturer(s)	:	Ts. DIYANA BT AB KADIR (FYP2 BCE COORDINATOR)
Submission Title	:	PROGRESS REPORT
Deadline	:	Day 15 Month MAY Year 2026 Time 11:50PM
Penalties	:	• 5% will be deducted per day to a maximum of four (4) working days, after which the submission will **not** be accepted. • Plagiarised work is an Academic Offence in University Rules & Regulations and will be penalised accordingly.

Section B: Academic Integrity

Tick (**√**) each box below if you agree:

√	I have read and understood the UniKL's policy on Plagiarism in University Rules & Regulations.
√	This submission is my own, unless indicated with proper referencing.
√	This submission has not been previously submitted or published.
√	This submission follows the requirements stated in the course.

Section C: Submission Receipt

(must be filled in manually)

Office Receipt of Submission

Date & Time of Submission (stamp)	Student Name(s)	Student ID(s)
22/5/2026	MD EJAJ MAHMUD	52222222123

Student Receipt of Submission

This is your submission receipt, the only accepted evidence that you have submitted your work. After this is stamped by the appointed staff & filled in, cut along the dotted lines above & retain this for your record.

Date & Time of Submission (stamp)	Course Code	Submission Title	Student ID(s) & Signature(s)
22/5/2026	IPB 49906	Progress Report 4	52222222123

UNIVERSITI KUALA LUMPUR
RUBRIC FOR PROJECT PROGRESS

COURSE CODE	IPB 49906	**TOTAL MARKS (50 MARKS)**
COURSE NAME	FINAL YEAR PROJECT 2	
COURSE SEMESTER/ YEAR	MARCH 2026	
STUDENT NAME	MD EJAJ MAHMUD	
STUDENT ID	52222222123	
ASSESSMENT DATE	WEEK 1 – WEEK 11	Smart Recycling: Reward-Based Reverse Vending Machine Using Proximity Sensor Detection for Plastic Waste Management
ASSESSMENT DUE	15 MAY 2026	
NAME OF SUPERVISOR	Dr. Grad. Engr. Hannah Binti Sofian	

NO.	CRITERIA	Description	WEIGHTAGE	SCORE 1-10	ACTUAL SCORE
1	**Consultation with Supervisor**	Meet Supervisor with significant progress	1		
2	**Timelines**	Record of research progress with respect to the Timelines	1		
3	**Research progress**	Show progress in research with related information and resources	1		
4	**Proposed Idea**	Ability to apply higher order thinking	1		
5	**Self-driven & Independency**	Ability to perform independently	1		
				TOTAL SCORE (50 MARKS)	

PROGRESS REPORT NO. 4

Student's Name: MD EJAJ MAHMUD

FYP Code : IPB49906

Semester	07		Week	09
Meeting No.	01		Date	15/05/2026

Progress (fill in by Student)

By Week 9, the main focus was hardware integration and functional testing. The Arduino Mega 2560, LCD, inductive proximity sensor, capacitive proximity sensor, IR sensor, ultrasonic sensor, LEDs, buzzer, and servo motor connections were assembled and checked on the prototype wiring setup.

The inductive and capacitive proximity sensors were tested using plastic bottle and metal/can objects. The plastic bottle test confirmed that the capacitive sensor could detect a PET bottle when it was placed close to the sensing area. The metal/can test confirmed that the inductive sensor responded correctly when a metallic object was introduced. This helped separate the acceptance condition from the rejection condition before final enclosure mounting.

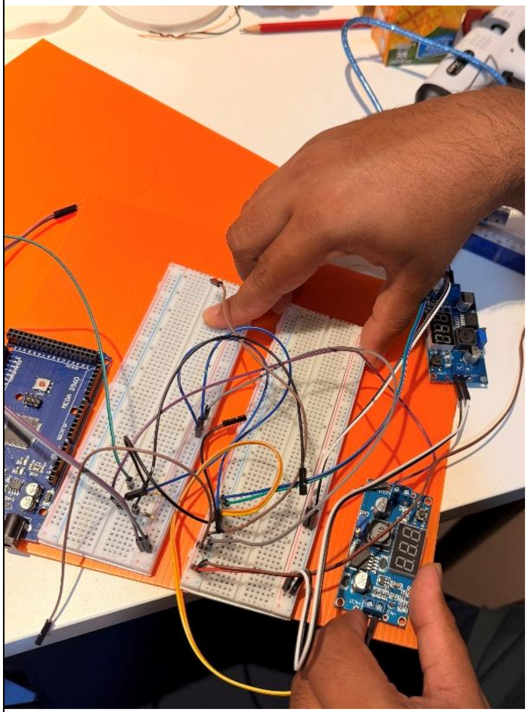
The LCD module was connected and tested to confirm that system messages could be displayed clearly to the user. This is important because the prototype depends on front-panel instructions to guide the user during bottle insertion and status feedback.

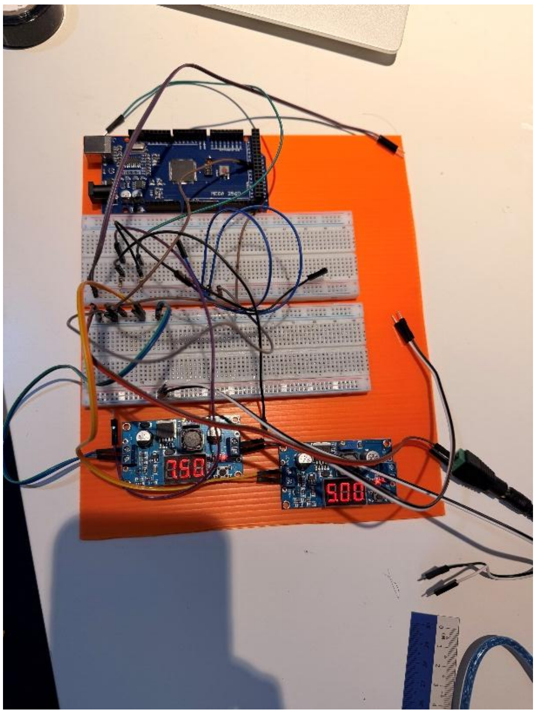
The prototype enclosure and wiring layout were also improved during this period. The bottle entry section, sensor placement area, and internal wiring arrangement were adjusted so that the circuit could be fitted into the physical structure. Some wiring was still temporary at this stage because the final mounting position of the sensors and servo mechanism needed further adjustment.

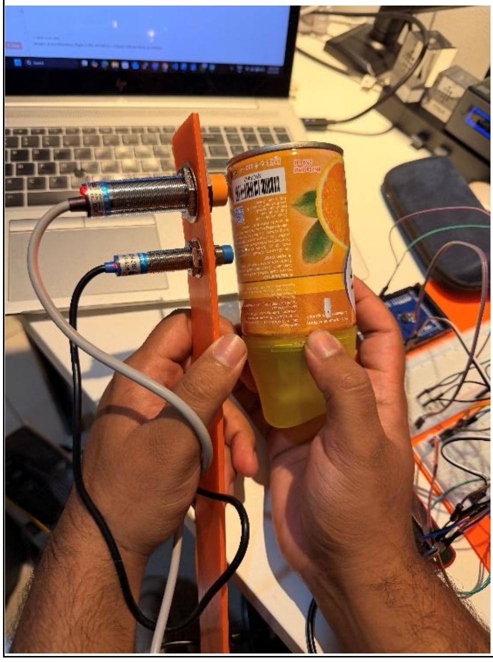
The main issue found during testing was sensor placement sensitivity. When the bottle was not close enough to the capacitive sensor, the reading could become inconsistent. To solve this, the sensor position was moved closer to the bottle path and the entry area was adjusted so the bottle passes through a more controlled position.

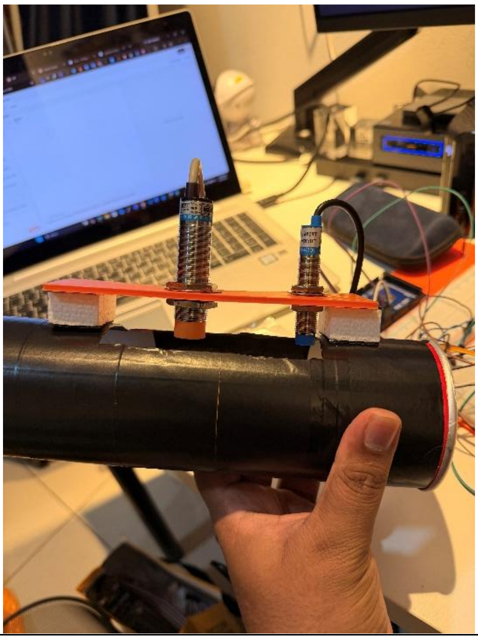
Based on the FYP2 timeline, this progress supports the Week 9 Progress Report 4 stage. The next work is to complete final prototype mounting, continue result testing, and prepare the final report, technical paper, and poster for supervisor review.

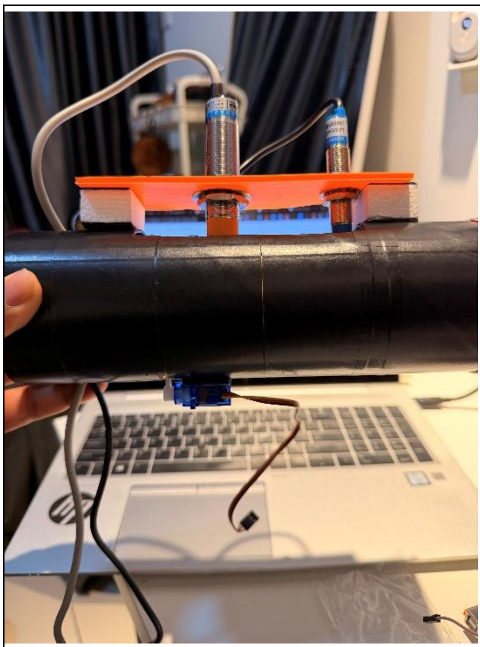
Appendix 1: Progress Report 4 Prototype Evidence.

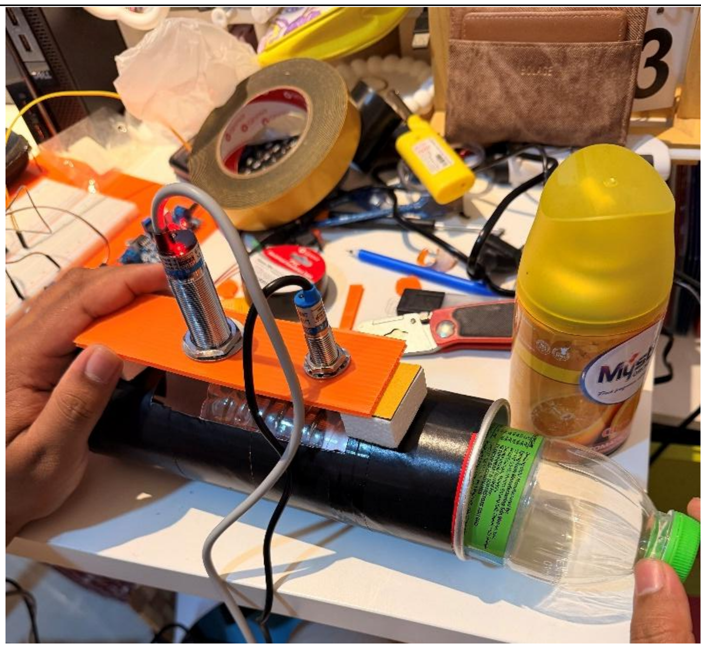












Comments (by Supervisor)	Signature and Name
ok good, please add IoT & AI as well as analysis for the project	DR HANNAH ST. SOFIAN